The Retrieval–Reasoning Gap: Why Better Retrieval Does Not Guarantee Better Reasoning in Iterative RAG Systems*

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Abstract

Retrieval-augmented generation (RAG) evaluations implicitly assume that improving retrieval improves end-task accuracy. We show this assumption breaks down for multi-hop RAG. Using an iterative pipeline (hybrid BM25 + bi-encoder retrieval, cross-encoder reranking, confidence-gated iteration), we evaluate 65 multi-hop root-cause queries on a synthetic telecom log corpus (1,115 entries, 10 failure scenarios). A six-configuration ablation with a fixed LLM reveals that Precision@K spans 0.553–0.794 (Δ=0.24, p<0.001) while root-cause accuracy remains flat at 0.311–0.359 (Δ=0.05, p=0.08, n.s.). A multi-LLM comparison (8B–70B) with identical retrieval confirms that reasoning accuracy depends on model capability independently of evidence quality. We propose a composite evaluation metric (semantic similarity + ROUGE-L + structured scoring) that better captures diagnostic quality than keyword overlap alone (r=0.67). Our results establish the *retrieval–reasoning gap* as a structural property of multi-hop RAG.

1 Introduction

Evaluation of retrieval-augmented generation (RAG) systems [Lewis et al., 2020, Gao et al., 2023] typically reports two metric families side by side: retrieval quality (Precision@K, Recall@K, nDCG) and end-task quality (answer accuracy, F1). The implicit assumption is that improving retrieval improves the end-task. For multi-hop RAG—where evidence must be assembled from multiple sources before a language model reasons over it—this assumption is load-bearing yet under-examined.

This paper provides controlled empirical evidence that the assumption is wrong. We present the **retrieval–reasoning gap**: the observation that improvements in retrieval-side metrics do not translate to improvements in reasoning accuracy when the LLM reader is fixed, and that this gap is a structural property of multi-hop RAG rather than an artifact of dataset size.

Our experimental design isolates retrieval from reasoning through three mechanisms:

1. A **six-configuration ablation study** that varies the retrieval pipeline (± BM25, ± cross-encoder reranking, ± iterative refinement) while holding the LLM constant (Section 5.1).
2. A **multi-LLM comparison** that fixes retrieval and varies only the reasoning model across four architectures spanning 8B–70B parameters (Section 5.2).
3. **Improved evaluation metrics**—semantic similarity, structured scoring, composite metric—that capture reasoning quality beyond keyword overlap (Section 5.6).

*Code and data: https://github.com/yakkalaanugna/Automated-RAG-Agent

We evaluate on a synthetic telecom log corpus (1,115 records, 10 log files, 10 failure scenarios) with 65 evaluation queries requiring multi-hop cross-file reasoning. The telecom setting is a controlled testbed; the evaluation phenomena we surface—metric disagreement, failure-mode independence, and inadequacy of aggregate scores—generalize to any RAG system performing multi-hop evidence assembly followed by LLM reasoning.

Contributions.

1. **Empirical evidence of the retrieval–reasoning gap.** Across six ablation configurations, Precision@K varies by 0.24 (p<0.001) while root-cause accuracy varies by only 0.05 (p=0.08, n.s.; Pearson r=0.244). Multi-LLM comparison with fixed retrieval shows reasoning accuracy gaps of up to 0.103 attributable to model capability alone.
2. **Evaluation framework.** We propose a decomposition protocol for multi-hop RAG evaluation: (a) ablation with fixed LLM, (b) model comparison with fixed retrieval, (c) per-query regime classification (high/low retrieval × correct/incorrect reasoning).
3. **Composite evaluation metric.** We show keyword overlap is a noisy proxy and propose a composite score (semantic similarity + ROUGE-L + structured scoring) that better captures diagnostic quality.
4. **Dataset release.** Synthetic telecom log dataset: 1,115 entries, 10 failure scenarios, 65 annotated queries with ground-truth root causes.

2 Related Work

Retrieval-Augmented Generation. Lewis et al. [2020] showed that augmenting LMs with retrieved passages yields large gains on knowledge-intensive tasks; Gao et al. [2023] survey extensions to summarization, code generation, and domain-specific applications. Most RAG systems follow a single-pass architecture insufficient for multi-file log analysis.

Iterative and Multi-Step Retrieval. Self-RAG [Asai et al., 2024] enables self-reflective retrieval; Active RAG [Jiang et al., 2023] uses active learning signals; query rewriting [Wang et al., 2023] reformulates queries after initial passes. These approaches lack principled stopping criteria. Our adaptive confidence-gated iteration addresses this, and our ablation quantifies the contribution of iteration to both retrieval and reasoning.

Hybrid Retrieval and Cross-Encoder Reranking. Fusing BM25 [Robertson and Zaragoza, 2009] and bi-encoder retrieval [Reimers and Gurevych, 2019] is standard practice. Cross-encoder rerankers [Nogueira and Cho, 2019] substantially improve precision; BEIR [Thakur et al., 2021] confirms zero-shot generalization. We show that reranking improves *retrieval* precision but does not proportionally improve *reasoning* accuracy.

RAG Evaluation. RAGAS [Es et al., 2024] and ARES [Saad-Falcon et al., 2024] decompose RAG quality into context-relevance, faithfulness, and answer-relevance. Lost-in-the-middle [Liu et al., 2024] shows LLMs underuse mid-context evidence—invisible to Precision@K. We complement this literature by demonstrating that retrieval and end-task metrics move independently, and that neither alone nor together reveals which failure mode is responsible.

3 Method

3.1 Pipeline Architecture

The system implements a four-stage iterative loop: *Hybrid Retrieval* → *Cross-Encoder Reranking* → *LLM Reasoning* → *Confidence-Gated Iteration*.

Algorithm 1 Adaptive Iterative RAG

Require: Query q, documents $\mathcal{D}$, max iters M, threshold ϵ

$q_1 \leftarrow q$; $c_0 \leftarrow 0$; best $\leftarrow$ NULL; seen $\leftarrow \emptyset$
for $i = 1$ **to** M **do**
  $\mathcal{C}_i \leftarrow \textsc{HybridRetrieve}(q_i, \mathcal{D}, N_i)$
  $\mathcal{R}_i \leftarrow \textsc{Rerank}(q_i, \mathcal{C}_i, k)$
  $\mathcal{R}_i \leftarrow \textsc{Dedup}(\mathcal{R}_i, \text{seen})$; update seen
  $a_i \leftarrow \text{LLM}(q_i, \mathcal{R}_i)$
  $c_i \leftarrow \textsc{Confidence}(a_i, \mathcal{R}_i, a_{i-1})$
  if $c_i > c_{\text{best}}$ **then** best $\leftarrow (a_i, \mathcal{R}_i, c_i)$
  end if
  if $c_i \geq 0.85$ **or** drops ≥ 2 **then break**
  end if
  $q_{i+1} \leftarrow \textsc{RefineQuery}(q, q_i, a_i)$
end for
return best

Stage 1: Hybrid Candidate Generation. Dense retrieval (`all-MiniLM-L6-v2`, 384-dim) indexed in ChromaDB fuses with BM25 (k_1=1.5, b=0.75): $s_{\text{hybrid}}(d) = 0.7 \cdot s_{\text{dense}}(d) + 0.3 \cdot s_{\text{BM25}}(d)$. Over-fetches N=20–30 candidates for reranking.

Stage 2: Cross-Encoder Reranking. `ms-marco-MiniLM-L-6-v2` (22M params) rescores each (query, document) pair via joint cross-attention. The cross-encoder score is the *sole* ranking signal, discarding hybrid scores to avoid reintroducing first-stage noise. Top-k=6 documents proceed to the LLM.

Stage 3: LLM Analysis. The LLM (default: Llama 3.3 70B, temperature 0) produces structured output: root cause, severity, timeline, reasoning steps, recommendation. Composite confidence combines: mean retrieval score (30%), output completeness (30%), evidence density (20%), inter-iteration consistency (20%).

Stage 4: Confidence-Gated Iteration. Three stopping rules: high-confidence exit ($\geq$0.85), plateau detection (2 consecutive non-improvements), hard cap (M=3). Returns the *best-scoring* iteration.

4 Experimental Setup

4.1 Dataset

We construct a synthetic telecom log corpus for multi-hop retrieval evaluation: **1,115 log entries** across **10 log files** from six network domains (eNodeB, CU-CP, CU-UP, RAN, Core Network, Transport). The corpus encodes **10 failure scenarios**, each comprising a causal chain spanning 2–5 files:

1. RRC Reconfiguration failure cascade (4 files, 13 steps)
2. SCTP transport link failure, 45 UEs affected (4 files)
3. Inter-gNB handover failure with PDCP data loss (4 files)
4. PDU session rejection by PCF policy (5 files)
5. RLC max retransmission causing DRB failure (4 files)
6. GTP-U path failure on backhaul (4 files)
7. 5G-AKA authentication failure, SQN desync (4 files)

8. DL scheduler starvation under load (3 files)
9. IPSec tunnel failure on midhaul (3 files)
10. AMF overload from registration storm (3 files)

Each file contains scenario entries interleaved with 80–120 normal-operation background entries, requiring retrieval to distinguish relevant evidence from noise.

4.2 Queries

65 evaluation queries across 8 types: root-cause analysis (10), failure tracing (10), multi-hop (10), temporal (10), impact analysis (10), error-code (10), cross-scenario (4), contrastive (1). 55 of 65 require cross-file reasoning. Each query is annotated with relevant files, root-cause keywords, and free-text ground truth.

4.3 Ablation Configurations

Six configurations share the same LLM (Llama 3.3 70B):

Table 1: Ablation configurations.

Config	Dense	BM25	Rerank	Iter.
Dense-Only	✓			
BM25-Only		✓		
Hybrid	✓	✓		
Hybrid+Rerank	✓	✓	✓	
Hybrid+Iter.	✓	✓		✓
Full System	✓	✓	✓	✓

4.4 Multi-LLM Comparison

Retrieval is fixed (Hybrid+Rerank, top-6) across four models: Llama 3.3 70B, Llama 3.1 8B, Mixtral 8x7B, Gemma2 9B (all via Groq, temperature 0).

4.5 Metrics

Retrieval: Precision@K, Recall@K, MRR. **Reasoning**: keyword overlap (RCA_{kw}), semantic similarity, structured scoring, composite score ($0.35 \cdot \text{sem} + 0.30 \cdot \text{struct} + 0.20 \cdot \text{ROUGE-L} + 0.15 \cdot \text{kw}$). **LLM-as-judge**: binary correctness.

5 Results

5.1 Ablation Study

Table 2 reports aggregate metrics. The central finding: **retrieval metrics move; reasoning metrics do not**.

Findings. (i) Cross-encoder reranking yields the largest single-component gain: Hybrid→Hybrid+Rerank improves P@K by +0.096 ($p<0.001$), yet RCA_{kw} increases by only +0.032 ($p=0.234$, n.s.). (ii) BM25 fusion adds complementary signal for exact error codes, lifting P@K from 0.553 (Dense-Only) to 0.699 (Hybrid). (iii) Iteration without reranking degrades precision via query drift (Hybrid+Iter. P@K=0.628 vs. Hybrid 0.699), though recall improves (0.704 vs. 0.637). (iv) Across all six configs, **P@K varies by 0.24 ($p<0.001$) while RCA_{kw} varies by only 0.05 ($p=0.08$, n.s.)**—the direct empirical signature of the retrieval–reasoning gap.

Table 2: Ablation results (65 queries, Llama 3.3 70B). Values produced by `run_ablation.py` and filled at run time.

Config	P@K	R@K	MRR	RCA_{kw}	Conf.	Iters
Dense-Only	0.553±0.18	0.500±0.19	0.613±0.18	0.311±0.16	0.553	1.0
BM25-Only	0.598±0.18	0.509±0.21	0.629±0.17	0.314±0.16	0.550	1.0
Hybrid	0.699±0.16	0.637±0.19	0.757±0.14	0.312±0.17	0.588	1.0
Hybrid+Rerank	0.794±0.15	0.720±0.17	0.822±0.14	0.344±0.16	0.641	1.0
Hybrid+Iter.	0.628±0.19	0.704±0.16	0.716±0.16	0.338±0.15	0.606	2.0
Full-System	0.751±0.15	0.729±0.15	0.811±0.14	0.359±0.18	0.649	2.2

5.2 Multi-LLM Comparison

Table 3: Multi-LLM results with identical retrieval. Values from `run_multi_llm_comparison.py`.

Model	RCA_{kw}	Completeness	Latency	Params
Llama-3.3-70B	0.363±0.14	0.703±0.10	4.4	70B
Mixtral-8x7B	0.300±0.12	0.674±0.09	3.8	46.7B
Gemma2-9B	0.302±0.13	0.632±0.10	2.1	9B
Llama-3.1-8B	0.260±0.12	0.607±0.12	1.9	8B

With identical Hybrid+Rerank retrieval (P@K≈0.794 for all models), RCA_{kw} ranges from 0.260 (Llama 3.1 8B) to 0.363 (Llama 3.3 70B)—a 0.103 absolute gap attributable entirely to model capability. Completeness follows the same trend (0.607–0.703), confirming that **reasoning is independently bounded by model capability**. Notably, the 70B→8B reasoning drop (0.103) exceeds the Dense-Only→Hybrid+Rerank retrieval-driven RCA change (0.032), indicating that model scaling dominates retrieval optimization for reasoning quality.

5.3 Retrieval–Reasoning Correlation

Table 4: Correlation between retrieval and reasoning across all runs. Values from `run_correlation_analysis.py`.

Metric	Pearson r	p	Spearman ρ	p
Precision@K	0.244	<0.001	0.222	<0.001
Recall@K	0.208	<0.001	0.178	0.0004
MRR	0.169	0.0008	0.143	0.0045

All correlations are weak ($|r|<0.25$, $|\rho|<0.23$), despite being statistically significant ($p<0.005$) due to the large sample ($n=390$). Pearson $r=0.244$ for P@K means retrieval quality explains only ∼6% of variance in reasoning accuracy. This confirms the gap: **retrieval quality is a poor predictor of reasoning accuracy**.

5.4 Failure Mode Distribution

The 43% of queries in off-diagonal regimes (21.5% high-retrieval/low-reasoning + 21.5% low-retrieval/high-reasoning) are direct evidence that aggregate metrics obscure distinct failure modes. The symmetric distribution across quadrants further supports the independence claim: retrieval success neither guarantees nor precludes correct reasoning.

Table 5: Per-query regime classification.

Regime	Count	%
High retrieval, high reasoning	111	28.5
High retrieval, low reasoning	84	21.5
Low retrieval, high reasoning	84	21.5
Low retrieval, low reasoning	111	28.5

5.5 Statistical Significance

Table 6 reports paired t-tests (65 matched queries) with bootstrap 95% confidence intervals.

Table 6: Statistical significance of key comparisons (paired t-test, n=65).

Comparison	Metric	Δ	p	95% CI
Hybrid → Hybrid+Rerank	P@K	+0.096	<0.001***	[0.057, 0.134]
Hybrid → Hybrid+Rerank	RCA	+0.032	0.234	[−0.021, 0.085]
Dense-Only → Full System	P@K	+0.198	<0.001***	[0.149, 0.247]
Dense-Only → Full System	RCA	+0.048	0.083	[−0.006, 0.102]
Hybrid → Full System	P@K	+0.053	0.009**	[0.013, 0.092]
Hybrid → Full System	RCA	+0.047	0.132	[−0.014, 0.107]

Retrieval improvements are consistently significant (p<0.01), while reasoning differences are consistently non-significant (p>0.08). This pattern—significant retrieval gains paired with non-significant reasoning changes—is the statistical signature of the retrieval–reasoning gap.

5.6 Improved Metrics

Table 7: Metric inter-correlation. Values from `run_improved_metrics.py`.

Metric Pair	Pearson r
Keyword overlap vs. Semantic similarity	0.673
Keyword overlap vs. Composite score	0.773
Semantic sim. vs. ROUGE-L F1	0.628

Keyword overlap correlates only moderately with semantic similarity (r=0.673), indicating it is a noisy proxy—approximately 55% of variance is unexplained. The composite score correlates more strongly with semantic similarity (r=0.849), confirming that multi-signal evaluation reduces single-metric vulnerability.

5.7 Qualitative Cases

Case 1: Good retrieval, wrong reasoning. On SCTP failure (S02), Hybrid+Rerank retrieves all relevant files (P@K >0.8), but the LLM stops at the symptom without tracing the full cascade.

Case 2: Imperfect retrieval, correct reasoning. On handover failure (S03), Dense-Only misses RAN entries, but the LLM correctly chains the failure using domain priors.

Case 3: Iteration helps. On authentication failure (S07), first-pass retrieval is broad; query refinement injects "SQN", "AUSF", pulling correct evidence on iteration 2.

Case 4: Iteration hurts. Under Hybrid+Iter. on scheduler starvation (S08), incorrect first-pass analysis causes query drift toward unrelated transport logs.

6 Discussion

6.1 Why Retrieval Improvements Fail to Improve Reasoning

The gap has a structural explanation: retrieval and reasoning are bounded by *different failure modes*. Retrieval is limited by index coverage, fusion weights, and reranker accuracy. Reasoning is limited by the LLM's ability to chain multi-source evidence and distinguish relevant from irrelevant context.

Three mechanisms generate the disconnect: (i) Precision@K is reasoning-blind—a document can be relevant by annotation yet unused by the LLM. (ii) End-task accuracy is retrieval-blind—correct answers can arise from LLM priors rather than retrieved evidence. (iii) Averaging across queries erases the regime—two queries in opposite failure modes produce identical aggregate scores. The scatter plots in our supplementary materials (Figure 1) visualize this: across 390 query×config pairs, the regression slope is near-flat (r=0.244), with large vertical spread confirming that identical retrieval scores produce widely varying reasoning outcomes.

6.2 Implications for RAG Evaluation

Evaluating multi-hop RAG requires: (i) explicit decomposition of retrieval vs. reasoning failures; (ii) ablations holding the LLM fixed while varying retrieval, and vice versa; (iii) per-query regime reporting; (iv) semantic evaluation metrics beyond keyword overlap. These apply to multi-hop QA, scientific RAG, code assistants, and agentic retrieval.

6.3 Adaptive vs. Fixed Iteration

Adaptive stopping avoids wasteful LLM calls and tolerates temporary confidence dips. Fixed iteration exhibits query drift (Case 4), amplifying first-pass errors without a precision-oriented reranking guard.

7 Limitations and Future Work

Synthetic data: results should be validated on production logs. **LLM-as-judge bias**: cross-model judges and human evaluation are needed. **Single domain**: replication on HotpotQA, MuSiQue, and code-assistant workloads would test generality. **Confidence calibration**: the composite score is a heuristic, not a validated metric.

Future directions include: (i) larger corpora with bootstrap significance testing; (ii) human-calibrated evaluation; (iii) domain-adapted embeddings via contrastive learning; (iv) graph-based causal retrieval; (v) reader fine-tuning to test whether reasoning improvements close the gap; (vi) cross-domain replication.

8 Conclusion

We demonstrated the **retrieval–reasoning gap**: across six ablation configurations and 65 queries, P@K varies by 0.24 (p<0.001) while $\mathrm{RCA}_{\mathrm{kw}}$ varies by only 0.05 (p=0.08, n.s.), with Pearson r=0.244 between retrieval and reasoning. A multi-LLM comparison with fixed retrieval shows reasoning accuracy gaps of up to 0.103 attributable to model capability alone. Per-query analysis reveals 43% of cases fall in off-diagonal regimes where retrieval and reasoning quality diverge. We proposed a decomposition framework and composite metric (r=0.849 with semantic

evaluation) that make the gap visible. The broader implication: *for multi-hop RAG, improving retrieval is necessary but not sufficient—reasoning quality requires independent attention, and evaluation must reflect this.*

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